Let's Talk About It: a NLP Approach to Evaluate Monetary Policy Communication

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Abstract

We develop a NLP pipeline to compute the *textual tone score index* of ECB monetary policy communication, quantifying the degree of hawkishness or dovishness. The pipeline leverages the RoBERTa Transformer model, and it is trained and evaluated on a sample of ECB press conferences following the policy rate announcement from 2000 to 2025. We show that the index allows to recover the correct theoretical effect of monetary policy on inflation, thus proving that the index meaningfully summarizes the central banks' evaluation of the underlying economic environment.

1 Introduction

"It is widely accepted that the ability of a central bank to affect the economy depends critically on its ability to influence market expectations about the future path of overnight interest rates" (Blinder et al., 2008, p.913). Therefore, central bank communication regarding monetary policy is a tool of utmost importance to achieve the desired policy goals. However, empirical evaluation of the impact of communication is complex, because the comprehension of human language is inherently nuanced and subjective.

We propose a methodology to translate transcripts of monetary policy communication into a quantitative measure suitable for econometric analysis. Specifically, we develop a Natural Language Processing (NLP) pipeline that takes as input the raw transcript of a central bank official statement and produces as output a *textual tone score*, that is a numerical quantification summarizing the degree of hawkishness or dovishness conveyed in the communication.

Our pipeline builds on the RoBERTa model (Liu et al. (2019)), a state-of-the-art Transformer architecture pre-trained on large text corpora to generate meaningful numerical representations of text. Intuitively, the model maps "human language" (i.e. words) into "machine language" (i.e. numbers) that can be used for downstream analytical tasks. Specifically, the pipeline consists of four steps. (I) It pre-processes the raw document and retains only economics-related sentences. (II) It identifies which of these sentences are specifically relevant to monetary policy. (III) It classifies the tone of each relevant sentence into five ordered categories: strongly dovish, dovish, neutral, hawkish, strongly hawkish. (IV) It aggregates sentence-level classifications into a single

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textual tone score for the document. Steps (II) and (III) rely on supervised learning. The authors independently annotated a corpus of sentences for monetary policy relevance and textual tone, and the Transformer models are trained on this labeled data to learn how to classify new sentences accordingly.

We apply this pipeline to a corpus of 264 ECB press conferences held after policy rate announcements between 2000 and 2025. Using these data, we construct a textual tone score index, i.e. a time series that tracks the evolution of communication tone over time. Our index exhibits a close correlation with the euro-area year-on-year HICP inflation rate and with the ECB deposit facility rate.

We show that our index meaningfully summarizes the central banks' evaluation of the underlying economic environment, because the index allows to recover the correct theoretical effect of monetary policy on inflation. Indeed, the index mitigates the endogeneity between policy actions and targets, as pointed out by the seminal work by Romer and Romer (2004), summarizing the central bank's evaluation of the economic environment.

The remainder of the paper is organized as follows. Section 2 reviews previous attempt to evaluate monetary policy communication. Section 3 describes data employed to implement and exploit our NLP pipeline. Section 4 explains in detail how the pipeline is assembled, trained, and executed. Section 5 exploits the textual tone score index to estimate the impact of monetary policy on inflation.

2 Related Work

The study of monetary policy communication has several dimensions both in terms of goals and tools. Some studies evaluate the tone of communication, other analyze authorship style, still others assess the similarity of statements. Techniques employed range from traditional text analysis methods to more advanced natural language processing approaches. In this section, we focus on studies that are most closely related to our application, namely those that aim to quantify the hawkish–dovish tone of central bank communication.

Among the simplest and most widely used approaches are *Bag-of-Words* (BoW) methods, which represent a document as a collection of word counts or frequencies and compute the tone based on the relative frequency of terms associated with tightening or easing monetary policy (Picault et al., 2022; Bennani and Neuenkirch, 2017; Baranowski et al., 2021). Despite their simplicity and interpretability, these approaches cannot properly capture context or semantic nuance, because they treat text as unordered collections of words or phrases. For example, a dictionary-based method that flags the bigram *"raise rates"* as hawkish would assign the same classification to the sentences *"we will raise rates"* and *"we will not raise rates"*, not taking into account the negation (Pfeifer and Marohl, 2023).

A major breakthrough to overcome these limitations came with the introduction of the Transformer architecture by Vaswani (2017), and the subsequent development of BERT (Bidirectional Encoder Representations from Transformers) by Google AI (Devlin, 2018). Transformer-based models generate bidirectional contextual embeddings, capturing word order, syntax, and long-range semantic relationships, so that the representation of a word varies depending on the sentence in which it appears. Transformer-based models have rapidly become the state-of-the-art in natural language processing and they have proven particularly well suited for capturing key aspects of policy communication.

Pfeifer and Marohl (2023) and Gambacorta et al. (2024) employ RoBERTa-based architectures to classify the monetary policy stance of central bank communication at the sentence level. The former trained a model on communication from several central banks for a binary classification task (positive vs. negative). The latter focused on Federal Open Market Committee statements and formulated a three-class classification problem (dovish, hawkish, neutral).

Other works go a step further and construct aggregate indices that measure the overall hawkishness–dovishness of an entire communication. Shah et al. (2023) develop a monetary policy stance index from FED documents based on three-class sentiment RoBERTa-large model.Lupton et al. (2023) and Lupton and Weitzenfeld (2024) employ five-class sentiment BERT-based models to assess the tone of communications by the ECB, the Federal Reserve, and the Bank of England. In a similar vein, Hallmark et al. (2024) compute the tone of ECB communications based on five-class RoBERTa model trained on Bloomberg news headlines.

The rapid rise of generative AI models was reflected in the study of monetary policy communication as well. Geiger et al. (2025) assess the sentiment of ECB communications, crafting prompts for the Llama 3.1 model. However, Gambacorta et al. (2024); Shah et al. (2023); Lupton and Weitzenfeld (2024) show that BERT-based model can still provide superior performance.

Our pipeline takes these contributions as a starting point for further improvement. We expand the dictionary of Shah et al. (2023) to identify economically relevant sentences in step (I). We assess the relevance of each sentence for monetary policy in step (II) following Lupton et al. (2023); Lupton and Weitzenfeld (2024). We prefer RoBERTa models over gen-AI options, because they ensure greater accountability, provide cost-effectiveness, and perform as well or better. We adopt a fine five-class sentiment classification in step (III) to capture the full gradation of policy stance, as in Lupton et al. (2023); Lupton and Weitzenfeld (2024); Hallmark et al. (2024). We develop a metric to address the ordinal and imbalanced nature of the data, inspired by Hallmark et al. (2024). We augment each sentence with information about the inflation environment to reduce ambiguity, as in Lupton et al. (2023); Lupton and Weitzenfeld (2024); Geiger et al. (2025).

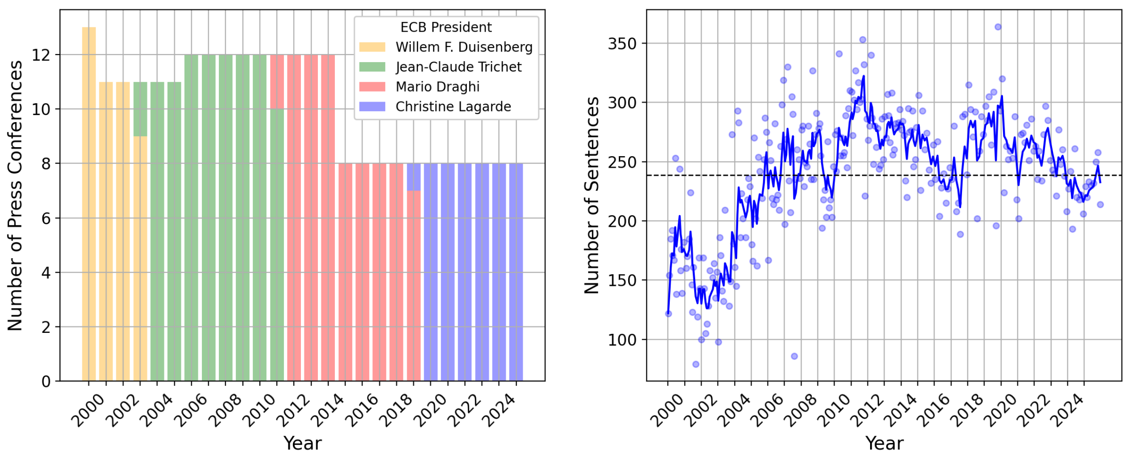
Using our tone index to estimate the effect of monetary policy on inflation is inspired by Romer and Romer (2004) and Aruoba and Drechsel (2026). The idea of purging monetary policy from the systematic component is endogenous to inflation can be traced back to the seminal contribution by Romer and Romer (2004). Their method relies on central bank forecasts to elicit the economic environment to which monetary policy is responding. Aruoba and Drechsel (2026) shows that central bank communication contains valuable information regarding the central bank's assessment of the economy on top of central bank's forecasts.

3 Dataset

The ECB Governing Council gathers at regular intervals to review and calibrate monetary policy conduct. After each decision, the ECB president holds a press conference, which lasts approximately 60 minutes and consists of two parts: an introductory statement to motivate the decisions and a Q&A session to interact with journalists. As shown in Figure 1(a), the number of press conferences varied over time, because the Council gathered roughly on a monthly basis before 2015, while meetings have been taking place approximately every six weeks since then. Transcripts were scraped directly from the ECB website.[1] As a result, we obtained a dataset containing 264 documents from 2000 to 2025; each document contains on average about 240 sentences, as indicated by the dashed

[1] `https://www.ecb.europa.eu/press/press_conference/monetary-policy-statement/html/index.en.html`.

black line in Figure 1(b). We retrieve from the ECB Statistical Data Warehouse the ECB policy rates and the quarterly euro area HICP inflation projections (series MPD.Q.U2.HIC.A); we retrieve from Bloomberg the monthly euro area HICP price index.



(a) ECB Press Conferences per Year

(b) Total Number of Sentences per Communication

Figure 1: Overview of the ECB press conference dataset. Fig.1(a) shows the number of ECB press conferences held each year from 2000 to 2025, bars are colored according to the ECB president in office. Figure 1(b) shows the number of sentences per press conference (scatter points) together with an exponentially weighted moving average. The dashed black line denotes the overall average length of ECB press conferences, equal to roughly 240 sentences.

4 Textual Tone Score Index

In this section, we describe the procedure to construct *textual tone score index*. The main steps are briefly summarized below and illustrated in Figure 2 and Figure **??**.

- *Step 1.* An initial *relevance filtering step*, in which we retain only economics-related sentences in ECB communications.
- *Step 2.* A *relevance classification model* that classifies each economics-related sentence as either relevant to monetary policy or not.
- *Step 3.* A *sentiment classification model* that assigns a sentiment (textual tone) label to each sentence.
- *Step 4.* The computation of the *textual tone score* for each ECB communication, defined as a weighted average of sentence-level textual tone scores using the relevance weights.

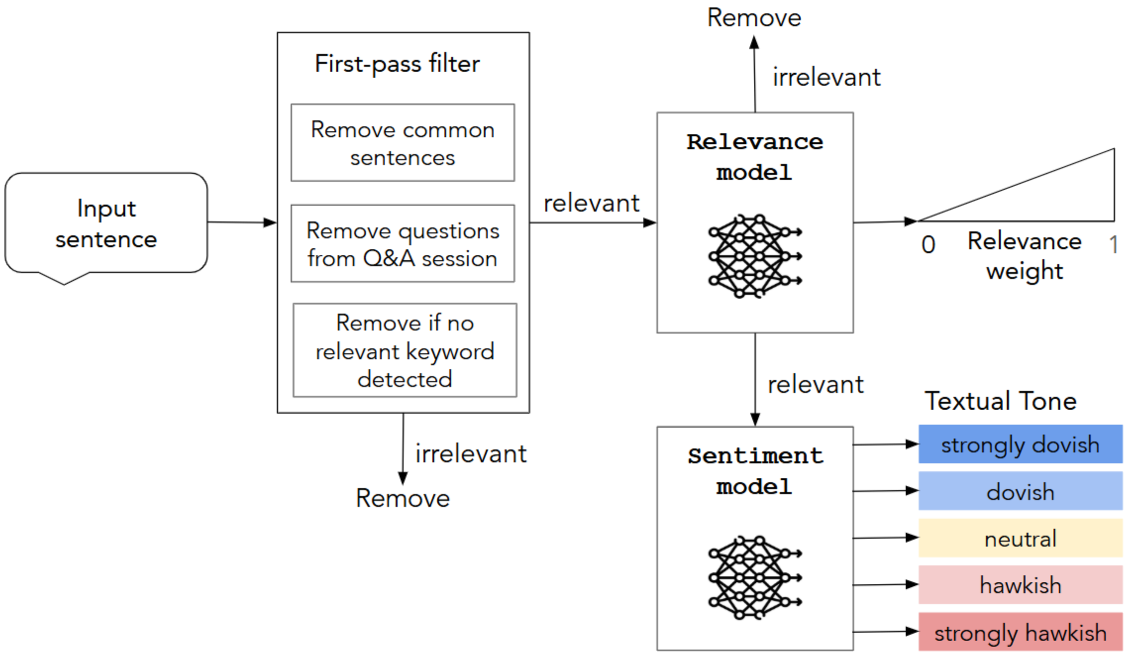


Figure 2: Overview of the NLP pipeline for sentiment classification.

4.1 Step 1: First-Pass Relevance Filtering

The purpose of this initial filtering step is to identify and remove economically irrelevant sentences, that is, sentences that are not useful for computing the textual tone score, such as greetings or questions from journalists. This step serves two main purposes. First, it improves computational efficiency by preventing these sentences from being passed to the more expensive classification models. Second, it reduces noise by removing completely irrelevant sentences, such as "*We are now ready to take your questions*", before the relevance classification step, ensuring that the *relevance model* only distinguishes between sentences that are relevant to monetary policy and those that are not. On average, 45% of the sentences in each document are removed (approximately 110 out of 240). The filtering procedure follows five steps described below and details are provided in appendix B:

1. *Sentence splitting.* We split the full document text (monetary policy statement and Q&A transcript) into sentences using "." and "?", which mark the end of a statement sentence or a question.

2. *Sentence merging using connector keywords.* We merge adjacent sentences when the second sentence begins with a *connector keyword*. We do this because the meaning of a passage may be distributed across multiple sentences. If treated separately, individual sentences may lose their meaning, because, for example, the second sentence might omit the subject and refer implicitly to the previous one.

3. *Removal of common sentences.* We remove overly common sentences, such as introductory greetings.

4. *Removal of journalists' questions.* We remove all questions asked by journalists during the Q&A session.

5. *Economics-related keyword filtering.* We apply a last filtering step by retaining only sentences that contain at least one *economics-related keyword* from a predefined dictionary, adapted from Shah et al. (2023) to better reflect ECB-specific terminology.

The effectiveness of these steps was validated through a series of manual checks on randomly selected samples of both removed and retained sentences, ensuring that the filtering procedure behaved as intended.

4.2 Preparatory Steps for Classification (Steps 2 and 3)

Before proceeding with the relevance and sentiment classification tasks, we first prepare the data for training and introduce the performance evaluation metrics used to assess model performance.

4.2.1 Data Preparation

For a classification task, we require a labeled dataset of the form

$$\mathcal{D} = \{(x_i, y_i),\ 1 \leq i \leq \mathcal{N}\}, \tag{1}$$

where x_i is the i-th sentence, y_i is the corresponding classification label used to train the models, and $\mathcal{N}$ is the total number of labeled elements in the dataset. Given the full set of document texts $\mathcal{X} = \{X_1, X_2, \ldots\}$, we then proceed with the *manual labeling* of sentences from a selected subset of documents in $\mathcal{X}$.

Two annotators assigned a value $y_i \in \{-2, -1, 0, 1, 2, 3\}$ to each sentence. The labels are defined as follows, and further details are provided in appendix C:

- *Label -2 (Strongly Dovish)*: Sentences that explicitly indicate a monetary policy action aimed at stimulating the economy and/or raising inflation, such as lowering interest rates or expanding the central bank's balance sheet. These statements reflect a strongly accommodative policy stance with a clear intention to act. For example,
  - "*Looking ahead, our monetary policy stance will remain accommodative for as long as necessary, in line with the forward guidance provided in July.*" (October 2, 2013)
  - "*In any case, we will conduct net asset purchases under the PEPP until the Governing Council judges that the coronavirus crisis phase is over.*" (June 4, 2020)

- *Label -1 (Dovish)*: Sentences that describe weak economic conditions, such as slow growth, high unemployment, or inflation below target. These reflect negative economic considerations, but do not indicate a concrete policy action. For example,
  - "*The risks surrounding the economic outlook for the euro area remain on the downside.*" (October 2, 2014)

  - "*Underlying price pressures in the euro area are expected to remain subdued over the medium term.*" (August 1, 2013)

- *Label 0 (Neutral)*: Sentences that are neither dovish nor hawkish, providing general information without suggesting a direction for monetary policy. For example,
  - "*Medium to long-term inflation expectations continue to be firmly anchored in line with price stability.*" (October 2, 2013)
  - "*The new Eurosystem staff projections show headline inflation averaging 2.1 per cent in 2025, 1.9 per cent in 2026, 1.8 per cent in 2027 and 2.0 per cent in 2028.*" (December 18, 2025)

- *Label 1 (Hawkish)*: Sentences that highlight strong economic conditions, rising inflation, or low unemployment. These reflect positive economic considerations, but do not explicitly signal a tightening action. For example,
  - "*The risks to the inflation outlook continue to be on the upside and have intensified, particularly in the short term.*" (July 21, 2022)
  - "*To sum up, annual inflation rates are projected to remain elevated in 2006 and 2007, with risks to this outlook remaining clearly on the upside.*" (November 2, 2006)

- *Label 2 (Strongly Hawkish)*: Sentences that explicitly indicate a tightening of monetary policy, such as raising interest rates, in response to high inflation or an economy growing faster than desired. These statements reflect a strongly restrictive policy stance with a clear intention to act. For example,
  - "*Our future decisions will ensure that the key ECB interest rates will be set at sufficiently restrictive levels for as long as necessary to achieve a timely return of inflation to our two per cent medium-term target.*" (July 27, 2023)
  - "*We took today's decision, and expect to raise interest rates further, to ensure the timely return of inflation to our two per cent medium-term inflation target.*" (October 27, 2022)

- *Label 3 (Irrelevant to Monetary Policy)*: Sentences that do not pertain to monetary policy and are included for training the relevance classification model. The initial relevance filter (Step 1) is not sufficient to exclude all sentences that, although related to the broader economy, are not specifically relevant to monetary policy. Therefore, an additional classification step is required to further distinguish between sentences that are useful for computing the tone (i.e., those directly related to monetary policy) and those that are not. For example,
  - "*To maintain the credibility of the fiscal policy framework it is essential to fully abide by its rules and to implement them in every respect.*" (October 2, 2003)
  - "*National fiscal and structural policies should aim at making the economy more productive and competitive, which would help to raise potential growth and reduce price pressures in the medium term.*" (July 18, 2024)

From the corpus of 264 press conference transcripts, we randomly selected four documents per year, corresponding to approximately 40% of the transcript corpus, for manual labeling as described above.

Before annotation, we applied the filtering procedure reported in Section 4.1 in order to exclude sentences that are completely irrelevant and will not be used in either the relevance or sentiment classification tasks. After filtering, the labeled dataset $\mathcal{D}$ consists of approximately 13,500 sentences. As illustrated in Figure 3, the class distribution appears roughly normal, with the extreme classes accounting for about 5% of the labeled dataset $\mathcal{D}$.

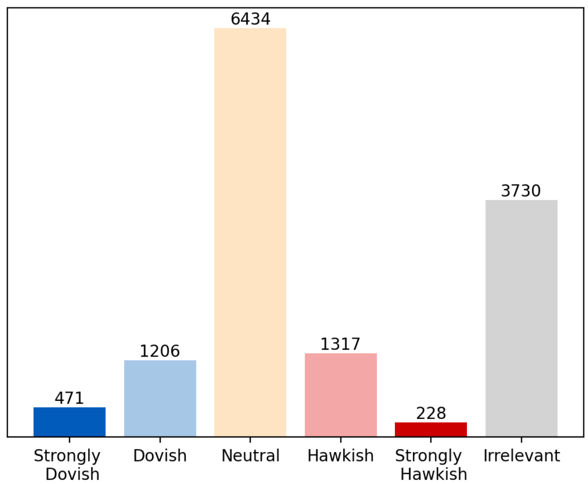


Figure 3: Distribution of sentence-level labels in the manually annotated dataset $\mathcal{D}$. The figure highlights the strong class imbalance, with the extreme labels (-2 and 2) accounting for only a small fraction of the observations.

Another issue concerns the *ambiguity* of certain sentences, whose interpretation depends on the prevailing economic context. This aspect was taken into account during manual labeling, as the annotators had access to the relevant macroeconomic information at the time of each press conference. However, a classification model does not have access to this contextual knowledge unless it is explicitly provided with additional information. For example, the sentence "*The Governing Council is determined to ensure that inflation returns to its 2% medium-term target in a timely manner*" can be interpreted as strongly hawkish when inflation is well above the target, but it may convey a dovish tone when inflation is below target.

To address this issue, we concatenate[2] to each sentence an inflation context label, which specifies the inflation environment with respect to the ECB inflation target. Initially, the ECB price stability mandate was articulated as a "medium-term inflation below 2%"; the target was first modified in May 2003 as "medium-term inflation below, but close 2%"; it was finally reviewed in July 2021 as "symmetric 2% medium-term inflation". Since the "medium term" evaluation horizon is inherently vague, we assign to each press conference the highest reading of the closest quarterly ECB HICP inflation forecasts for the next four quarters, $\pi^{forecast}_{Q1:Q4}$, (series MPD.Q.U2.HIC.A). The inflation context label states whether this inflation reading is below, at or above the prevailing target, which

[2]The context label c_i is then concatenated to the sentence text x_i to form an augmented input $x^c_i = x_i \ \langle/s\rangle \ c_i$. This results in an enriched labeled dataset $\mathcal{D}^c = \{(x^c_i, y_i), \ 1 \leq i \leq N\}$.

is specified as follows and results in the inflation target time line displayed in figure 4.

$$\text{Inflation Target} = \begin{cases} \pi_{Q1:Q4}^{forecast} < 2.00\% & \text{Before May 2003,} \\ 1.80\% \leq \pi_{Q1:Q4}^{forecast} < 2.00\% & \text{Between May 2003 and July 2021,} \\ \pi_{Q1:Q4}^{forecast} = 2.00\% & \text{After July 2021} \end{cases}$$

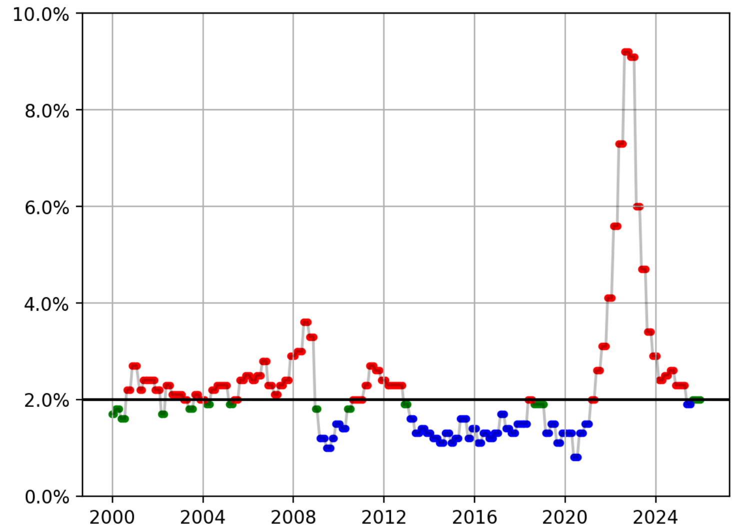


Figure 4: Inflation timeline used to construct the context labels, which help resolve sentiment ambiguity in sentence classification. Red dots represent an inflation reading above the target, green at the target, blue below the target.

The final step in preparing the data for training is randomly splitting the manually labeled dataset into training, validation, and test sets.[3] Given the strong class imbalance, we apply a stratified split to ensure that each subset preserves the same class proportions as the full dataset $\mathcal{D}^c$. This prevents the validation and test sets from containing too few observations from minority classes, such as -2 and 2. We denote the training, validation, and test sets as $\mathcal{D}^c_{\text{Tr}}$, $\mathcal{D}^c_{\text{V}}$, and $\mathcal{D}^c_{\text{Te}}$, respectively.

4.2.2 Performance Evaluation Metrics

We require an evaluation metric to assess the performance of a model, that is, its ability to correctly classify observations and to generalize to new and unseen data drawn from the same distribution. For classification tasks, a standard performance metric is *accuracy*, which is defined as

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(\hat{y}_i = y_i), \tag{2}$$

[3]The training set (in-sample) is used to train the model and learn patterns from the data. The validation set (out-of-sample) is used for model selection and hyperparameter tuning, helping to prevent overfitting to the training data. The test set (out-of-sample) is reserved for an unbiased evaluation of the final model's performance.

Number of press conference texts	104
Share of press conference texts	40%
Number of labeled sentences	≈13,500
Total number of labels	6
Number of sentiment labels	5

Table 1: Overview of the dataset used for training and evaluation.

where y_i and $\hat{y}_i$ denote the true and predicted labels for observation x_i, respectively, and $\mathbb{I}(\cdot)$ is the indicator function. However, as discussed in Section 4.2.1, the class distribution in our dataset is highly imbalanced, with the extreme sentiment classes accounting for only about 5% of the dataset, while the neutral class represents approximately 50%. Standard accuracy assigns equal weight to each observation rather than to each class, implying that errors in minority classes contribute less to the overall score. As a result, accuracy tends to favor models that perform well on majority classes at the expenses of rarer but important categories (such as -2 and 2). To address this issue, we can adopt *balanced accuracy*, which assigns equal importance to each class by averaging class-wise recall.[4] In the multiclass case, balanced accuracy is defined as

$$\text{Balanced Accuracy} = \frac{1}{\mathcal{C}} \sum_{k=1}^{\mathcal{C}} \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k}, \tag{3}$$

where $\mathcal{C}$ is the number of classes, and TP_k and FN_k denote the number of true positives and false negatives for class k, respectively. This metric ensures that performance on minority classes contributes equally to the final evaluation, making it more appropriate for our imbalanced classification setting. To interpret the accuracy of a model we can compare it to the baseline accuracy. For binary classification, the baseline accuracy is 50%, while for a 5-class classification problem, the baseline is 20%. Any accuracy above the baseline shows an improvement over random guessing.

Another important consideration is that sentiment classes are *ordered* rather than nominal:

$$\text{“strongly dovish”} < \text{“dovish”} < \text{“neutral”} < \text{“hawkish”} < \text{“strongly hawkish”}.$$

Consequently, not all misclassifications are equally severe. For example, classifying a *strongly dovish* (-2) sentence as *dovish* (-1) is substantially less problematic than classifying it as *strongly hawkish* (2). This introduces an *ordinality* issue in addition to class imbalance, implying that an appropriate evaluation metric should penalize classification errors more heavily the larger the distance between y_i and $\hat{y}_i$ in the ordered scale. A commonly used metric for ordinal classification problems is the *Quadratic Weighed Kappa* (QWK). It measures the level of *agreement* between two raters—here, the human annotators and the classification model—who classify items to an ordered set of categories. This measure ranges from -1 (complete disagreement) to 1 (perfect agreement), with 0 indicating agreement no better than random. In our setting, higher values of QWK correspond to better classification performance, with 1 representing perfect classification. The QWK is defined as:

$$QWK = 1 - \frac{\sum_{i,j} W_{i,j} O_{i,j}}{\sum_{i,j} W_{i,j} E_{i,j}} \tag{4}$$

[4] Also known as True Positive Rate (TPR).

where

- O is the *observed confusion matrix* comparing predicted and true labels. It is a $\mathcal{C} \times \mathcal{C}$ matrix, where diagonal elements correspond to normalized counts of correctly classified observations and off-diagonal elements to misclassifications.

- W is the *weight matrix*, whose entries quantify the penalty associated with each type of disagreement. Each element is defined as

$$W_{i,j} = \frac{(i-j)^2}{(\mathcal{C}-1)^2}. \tag{5}$$

  The diagonal entries are equal to zero (when $i = j$), reflecting the absence of penalty for correct classifications. The weights lie in the interval $[0, 1]$ and increase with the severity of the error, measured by the distance between the predicted class i and the true class j on the ordinal scale. When $\mathcal{C} = 5$, the possible penalty values are $\{0, 0.0625, 0.25, 0.5625, 1\}$, with the best-case scenario corresponding to $i = j$ and the worst-case scenario occurring when $|i - j| = \mathcal{C} - 1$.[5]

- E denotes the *expected confusion matrix* under chance agreement, computed by preserving the marginal class distributions.

While QWK is more robust than standard accuracy in the presence of class imbalance, it is not explicitly designed to address it. To account for both ordinality and class imbalance, we introduce a balanced variant of this metric, which we refer to as *Balanced Quadratic Weighted Kappa* (BQWK). The BQWK is computed by evaluating the QWK separately for each class and then averaging the resulting values across the $\mathcal{C}$ classes. This ensures that each class contributes equally to the final score, independently of its frequency, while preserving the ordinal structure of the sentiment labels.

4.3 Step 2: Relevance Model

After the first-pass filtering and data preparation steps, we proceed to the next stage of the NLP pipeline, namely *relevance classification*, where sentences are classified as either relevant to monetary policy or not.

We begin by considering the context-augmented labeled dataset $\mathcal{D}^c$ and removing duplicate sentences, as several sentences recur across different monetary policy statements. After deduplication, the dataset contains approximately 12899 observations. Next, we remap the original labels y_i to construct a *binary relevance target*. Specifically, 2911 sentences originally labeled as 3 are mapped to 0, while the 7408 remaining labels associated with sentiment values from -2 to 2 are mapped to 1, indicating relevance to monetary policy. Finally, we split the dataset into training, validation, and test sets using an 80%–10%–10% split, obtaining the datasets $\mathcal{D}_{\mathrm{Tr}}$, $\mathcal{D}_{\mathrm{V}}$, and $\mathcal{D}_{\mathrm{Te}}$, with sizes 10319, 1290, and 1290, respectively. The split is performed in a stratified manner to preserve the same class proportions across all subsets (Sadaiyandi et al. (2023)).

With the dataset prepared, we proceed to the classification stage. We use the Transformer-based model *RoBERTa*[6] (see Appendix A.2 for an overview of Transformers and the RoBERTa

[5] Labels are mapped to integer indices as $-2 \mapsto 0,\ -1 \mapsto 1,\ 0 \mapsto 2,\ 1 \mapsto 3,\ 2 \mapsto 4$.

[6] `https://huggingface.co/FacebookAI/roberta-base`. Two versions are available, base and large, differing in the number of parameters. We found that the base version performs slightly better for the relevance classification task.

architecture), implemented via the `transformers` library from Hugging Face. Specifically, we adopt the base version of the model, which consists of 12 layers and 125 million parameters.[7] We load both the pretrained RoBERTa model and its corresponding tokenizer.[8] The tokenizer is applied to the sentences in the training, validation, and test sets, with a maximum sequence length of 256 tokens. Sentences exceeding this length (which are rare) are truncated, while shorter sentences are padded to ensure a uniform input length across all inputs.[9]

The RoBERTa model is pretrained on large-scale, general-purpose text corpora, but it must be adapted to our specific dataset and to the binary relevance classification task. This adaptation is performed through *fine-tuning* (see Appendix A.5 for details), which consists of retraining a subset of the model parameters using a small learning rate. Training is carried out using *mini-batch gradient descent*. At each iteration, a mini-batch of 32 sentences is randomly sampled from the training set $\mathcal{D}_{\mathrm{Tr}}$ and used to compute the gradient of the loss function and update the model parameters. We optimize a *weighted cross-entropy loss* function to account for class imbalance. For a binary classification problem with classes $\mathcal{C} = \{0, 1\}$ and a mini-batch $\mathcal{B}$, the loss is defined as

$$\mathcal{L} = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \left[w_1 \cdot y_i \log(\hat{p}_i) + w_0 \cdot (1 - y_i) \log(1 - \hat{p}_i) \right], \tag{6}$$

where $y_i \in \{0, 1\}$ is the true label, $\hat{p}_i$ is the predicted probability that sentence x_i belongs to class 1, and w_1 and w_0 are the class weights for the positive (relevant) and negative (not-relevant) classes. The weights are computed as[10]

$$w_c = \frac{|\mathcal{D}_{Tr}|}{|\mathcal{C}| \cdot |\{i \in \mathcal{D}_{Tr} : i = c\}|}, \quad c \in \mathcal{C}. \tag{7}$$

In our setting, the relevant class ($c = 1$) is the majority class, with weight

$$w_1 = \frac{10319}{2 \cdot 7408} \approx 0.70,$$

while the not-relevant class ($c = 0$) is the minority class, with weight

$$w_0 = \frac{10319}{2 \cdot 2911} \approx 1.77.$$

Assigning a higher weight to the minority class increases the penalty associated with misclassifications of not-relevant sentences, thus mitigating the effects of class imbalance during training.

The training procedure also requires the selection of several *hyperparameters*, which can heavily influence model performance. The hyperparameters we consider are:

- Number of epochs, set to 10.
- Learning rate, for which we tested three values: $1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}$.

[7] We also experimented with the large version of RoBERTa, but it did not yield measurable performance improvements, so we retained the smaller model.

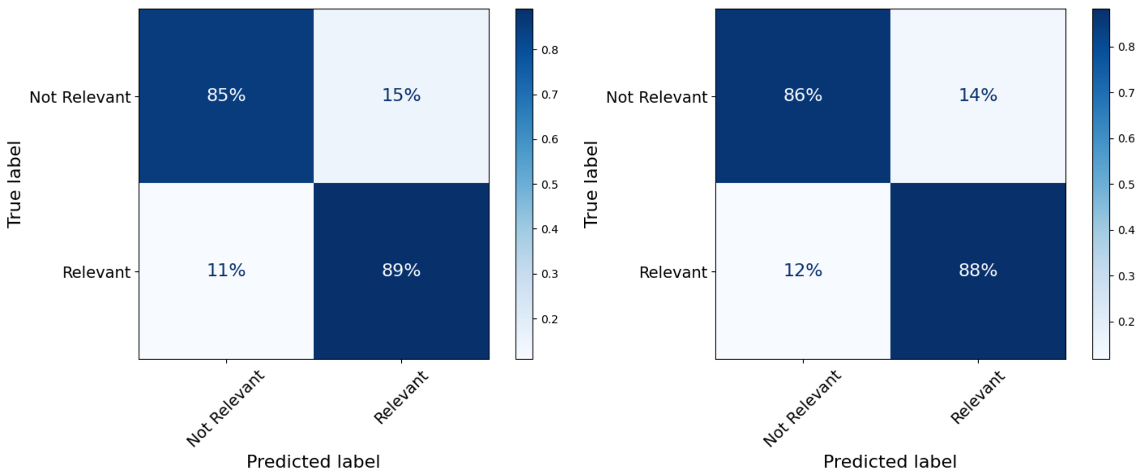
[8] The tokenizer converts the sentence text into tokens, that is, basic textual units such as words or subwords.

[9] Padding consists of adding a special padding token with value 1 to the end of shorter sequences so that all inputs have the same length, which is required for efficient batch processing in Transformer models.

[10] The class weights are computed as in `sklearn.utils.class_weight.compute_class_weight`.

- Number of frozen layers l, i.e., layers whose weights remain fixed while the last $12 - l$ layers are fine-tuned. We tested $l \in \{0, 3, 6, 9\}$.

We train the model on all combinations of learning rate and frozen layers to identify the best configuration. The *best model* is selected as the one achieving the highest balanced accuracy on the validation set $\mathcal{D}_V$, which contains data unseen during training. To prevent overfitting, we implement *early stopping* with a *patience* of 3 epochs. This means that if the balanced accuracy on the validation set does not improve for 3 consecutive epochs, training is stopped and the model weights corresponding to the highest validation accuracy are restored.



(a) Validation Set Confusion Matrix

(b) Test Set Confusion Matrix

Figure 5: Classification performance of the RoBERTa-base relevance model. The left panel shows the confusion matrix on the validation set, while the right panel reports the confusion matrix on the test set.

The best relevance model is obtained with 0 frozen layers and a learning rate of 1×10^{-5}, achieving a balanced accuracy of 87.13% on the test set. The model is then evaluated on the test set, containing entirely unseen data, yielding a balanced accuracy of 87.25%, which demonstrates strong generalization. Figure 5 shows the confusion matrices for the validation and test sets.

4.4 Step 3: Sentiment Model

Following the relevance classification model, the next step in the NLP pipeline is *sentiment classification*, where sentences that are relevant to monetary policy are assigned to one of five sentiment classes, as described in Section 4.2.1. We begin by considering the context-augmented dataset $\mathcal{D}^c$ and remove duplicate sentences. We then retain only sentences with sentiment labels ranging from -2 to 2, discarding those labeled as 3. To facilitate model training, we remap the original sentiment labels according to the transformation

$$-2 \mapsto 0, \ -1 \mapsto 1, \ 0 \mapsto 2, \ 1 \mapsto 3, \ 2 \mapsto 4.$$

As discussed previously, the sentiment classes are strongly imbalanced, with approximately 70% of sentences belonging to the neutral class and only about 7% belonging to the extreme classes (*strongly dovish* and *strongly hawkish*). The dataset is then split into training, validation, and test sets using a 70%–15%–15% split, performed in a stratified manner to preserve class proportions across all subsets. This results in the training, validation, and test datasets $\mathcal{D}_{Tr}$, $\mathcal{D}_{V}$, and $\mathcal{D}_{Te}$, with sizes 6482, 1389, and 1389, respectively.

For this task we experimented with different variants of the Transformer-based RoBERTa model,[11] and, unlike the relevance classification task, we employ the *large* version of RoBERTa, which consists of 24 layers and approximately 355 million parameters. In contrast to the relevance task, the larger model yields better performance for sentiment classification, likely due to the greater complexity of the task, which benefits from increased model capacity. After loading the model and its tokenizer, we tokenize all sentences in the training, validation, and test sets, using a maximum sequence length of 256 tokens, as in the relevance classification step.

The model is fine-tuned on the sentiment dataset and adapted to the new task using mini-batch gradient descent. At each training iteration, a mini-batch of 32 sentences is randomly sampled from the training set and used to update the model parameters. We optimize a *weighted cross-entropy loss* function to account for class imbalance.[12] For a 5-class classification problem with classes $\mathcal{C} = \{0, 1, 2, 3, 4\}$ and a mini-batch $\mathcal{B}$, the weighted cross-entropy loss is defined as

$$\mathcal{L} = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \sum_{c \in \mathcal{C}} w_c \cdot \mathbb{I}\{y_i = c\} \cdot log\left(\hat{p}_{i,c}\right), \tag{8}$$

where $\hat{p}_{i,c}$ is the predicted probability that sentence x_i belongs to class c, y_i is the ground truth label of sentence x_i, w_c is the class-specific weight, and $\mathbb{I}\{y_i = c\}$ is an indicator function equal to 1 if $y_i = c$ and 0 otherwise. This loss function is a direct generalization of the one used for the relevance model to the multiclass setting. Class weights are computed analogously in order to account for class imbalance. Specifically,

$$w_0 = \frac{6482}{5 \cdot 305} = 4.25, w_1 = \frac{6482}{5 \cdot 812} = 1.60, w_2 = \frac{6482}{5 \cdot 4340} = 0.30, w_3 = \frac{6482}{5 \cdot 897} = 1.45, w_4 = \frac{6482}{5 \cdot 128} = 10.13.$$

As expected, minority classes receive substantially larger weights, ensuring that misclassifications in these classes contribute more heavily to the loss and mitigating the strong class imbalance present in the data.

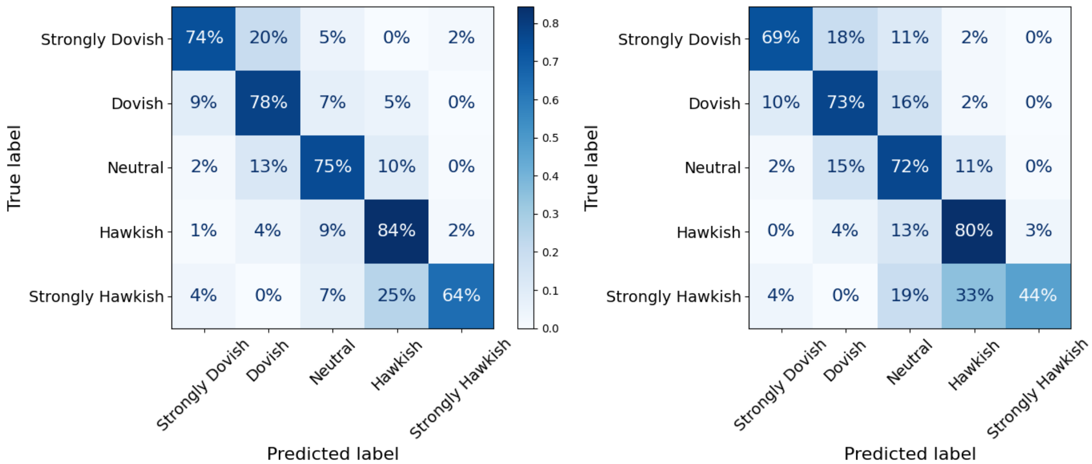
The hyperparameters that we consider are:

- Number of epochs, set to 30.
- Learning rate, for which we tested three values: $1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}$.
- Number of frozen layers l, i.e., layers whose weights remain fixed while the remaining $24 - l$ layers are fine-tuned. We tested $l \in \{0, 3, 6, 9\}$.

[11] We also experimented with the central-bank language models (CB-LMs) proposed by Gambacorta et al. (2024), which are RoBERTa models further pre-trained on a large corpus of central bank speeches, policy documents, and research papers. However, after fine-tuning on our labeled dataset, these models performed worse on the sentiment classification task than the standard RoBERTa-large model.

[12] We also tried the CORN loss `https://raschka-research-group.github.io/coral-pytorch/api_subpackages/coral_pytorch.losses/`, which explicitly accounts for class ordinality as discussed in Section 4.2.2, but we did not observe performance improvements.

We train the model on all combinations of learning rate and frozen layers to identify the best configuration. The *best model* is selected as the one achieving the highest balanced accuracy on the validation set $\mathcal{D}_V$. To prevent overfitting, we implement *early stopping* with a *patience* of 4 epochs,[13] and restore model weights corresponding to the highest validation accuracy.



(a) Validation Set Confusion Matrix

(b) Test Set Confusion Matrix

Figure 6: Classification performance of the RoBERTa-large sentiment model. The left panel shows the confusion matrix on the validation set, while the right panel reports the confusion matrix on the test set.

The best-performing sentiment classification model is obtained with 15 frozen layers and a learning rate of 3×10^{-5}, achieving a BQWK score of 75.21% on the validation set and 72.34% on the test set. Figure 6 reports the corresponding confusion matrices for the validation and test datasets. Importantly, the upper-right and lower-left regions of the matrices are largely empty, suggesting that severe misclassifications, such as labeling a *strongly dovish* sentence as *strongly hawkish*, are rare. This pattern indicates that the inclusion of context labels is effective in mitigating extreme ordinal errors.

4.5 Step 4: Textual Tone Score Index Computation

Now that the main components of the NLP pipeline have been introduced, the final step is to combine them into a single end-to-end procedure. For each press conference, we feed the full text of the introductory statement together with the transcript of the Q&A session into the pipeline. As a result, for each press conference we obtain a label for every sentence in the text. Sentences that are relevant to monetary policy are assigned a sentiment score ranging from -2 to 2, while all remaining sentences are assigned the label 3 indicating irrelevance for monetary policy.

Figure 7 reports the raw counts of sentence-level sentiment labels over time. Pronounced peaks in *hawkish* and *strongly hawkish* sentences between 2004 and 2008 reflect the global financial crisis

[13] We set the patience to four epochs, one more than in the relevance classification task, to account for the greater complexity of the sentiment classification problem. Given the larger number of classes and the more nuanced distinctions between them, the model may require additional training time to converge to a stable optimum.

build-up, while elevated values around 2011–2012 correspond to the post-crisis recovery. The most prominent increase in hawkish language occurs in 2022–2023, coinciding with the sharp rise in inflation. Conversely, peaks in *dovish* and *strongly dovish* sentences are observed during periods of economic stress, most notably during the great financial crisis and the subsequent adoption of non-conventional policy measures, with the strongest peak occurring during the Covid-19 pandemic, reflecting a more accommodative monetary policy stance.

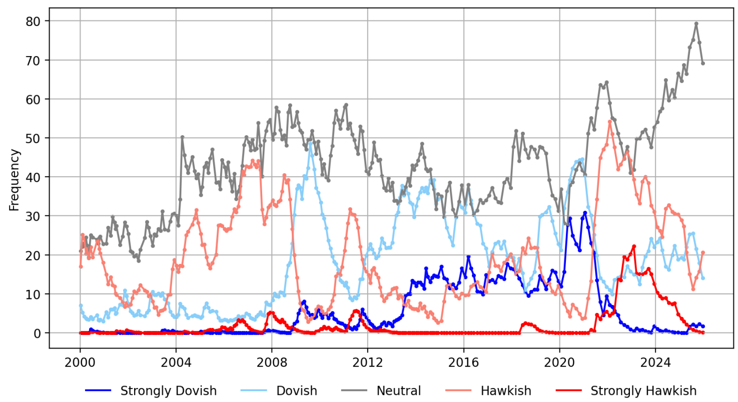


Figure 7: Raw counts of sentence-level sentiment labels, ranging from *strongly dovish* to *strongly hawkish*, for each press conference.

Based on these sentence-level classifications, we compute a single *textual tone score* for each press conference, which summarizes the overall dovish–hawkish tone of the communication. The tone score for the i-th press conference is defined as

$$t_i = \frac{1}{\mathcal{N}} \sum_{j=1}^{\mathcal{N}} w_j^{rel} \cdot \hat{y}, \tag{9}$$

where $\mathcal{N}$ is the total number of relevant sentences in the press conference, $\hat{y}_j$ is the sentiment label predicted for sentence j, and w_j^{rel} is a *relevance weight* derived from the relevance classification model's *confidence score*.[14]

We compute the textual tone score for each of the 264 press conferences held between January 2000 and December 2025. In theory, the score ranges from -2 to 2, corresponding to the extreme case in which all monetary policy-relevant sentences receive a relevance weight of 1 and are classified as either *strongly dovish* or *strongly hawkish*. In practice, the observed range is considerably narrower, spanning from -1.05 (April 30, 2020), the most dovish communication in the sample, to 0.67 (December 15, 2022), the most hawkish one.

The resulting textual tone index is plotted against context HICP inflation (cfr. figure 4) in figure 8 and against the Deposit Facility Rate (DFR) in figure 9. Overall, the index co-moves with both inflation and the policy rate, suggesting that the constructed textual tone measure captures meaningful variations in the monetary policy stance over time. Focusing on the policy rate, a preliminary visual inspection shows that changes in the communication tone anticipate variations of the policy rate, implying that communication prepares the ground for monetary policy action.

[14] The confidence score reflects the model's estimated probability that a sentence is relevant for monetary policy.

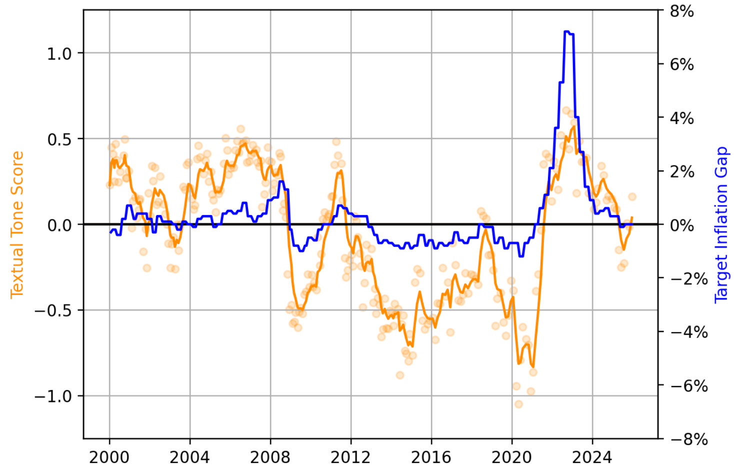


Figure 8: Textual tone score and context inflation gap. The orange line represents the exponentially weighted moving average of the tone score with a half-life of 60 days, while the orange dots denote the raw tone scores computed for individual press conferences. The blue line shows the context inflation gap relative to the 2% target.

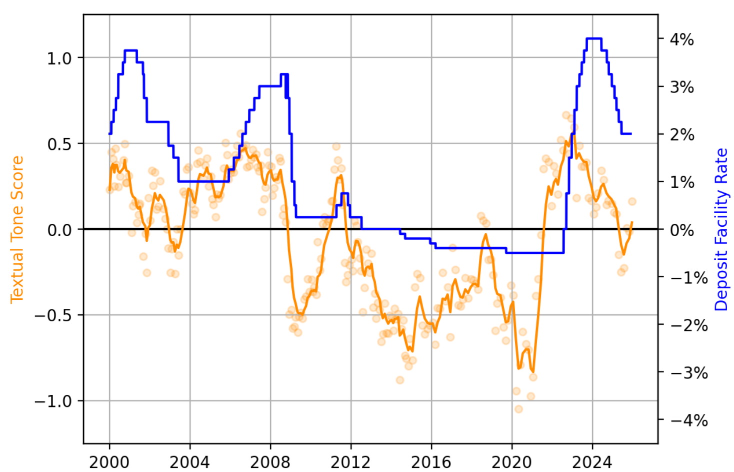


Figure 9: Textual tone score and Deposity Facility Rate (DFR). The orange line represents the exponentially weighted moving average of the tone score with a half-life of 60 days, while the orange dots denote the raw tone scores computed for individual press conferences. The blue line shows DFR.

We next examine the textual tone score at a topic-specific level, focusing on four main dimensions: *monetary policy*, *inflation*, *economic growth*, and *financial conditions*. This analysis allows us to compare how the tone associated with different topics evolves over time and to point out potential discrepancies across thematic dimensions. For each topic, we define a set of keywords that identify the corresponding thematic content. Each sentence in the press conferences is then associated with one or more topics based on the presence of these keywords. The keyword dictionaries contain 170 entries for monetary policy, 108 for inflation, 409 for growth, and 224 for finance.[15] We compute the topic-specific textual tone scores in the same manner as the overall tone score, but restricting the calculation to sentences that contain at least one keyword associated with the given topic.

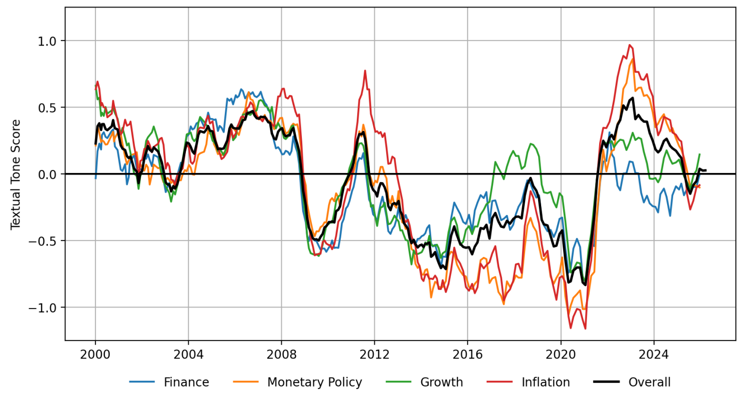


Figure 10: Comparison of textual tone scores associated with different topic categories. The black line represents the overall textual tone index, while the blue, orange, green, and red lines report the tone scores associated with financial, monetary policy, growth, and inflation-related content, respectively.

Figure 10 displays the four topic-specific tone indices together with the overall tone index. It can be noted that after 2022 communication turned hawkish with regard to inflation and monetary policy, while growth received a more moderate assessment. This is consistent with the utmost urgency of the fight against extreme inflationary pressures in the context of the fragile post-Covid recovery. Specularly, between 2016 and 2020 communication turned extremely dovish with regard to inflation and monetary policy, while growth assessment improved. This is consistent with the systematic deployment of unconventional monetary policies to stabilise inflation at the target amid improvements in the growth outlook. Overall, it can be noted that monetary policy and inflation broadly moved together, reflecting the price stability mandate of the ECB.

5 The Impact of Monetary Policy: a Macroeconomic Application

Studying the impact of monetary policy on the economy requires confronting with the endogeneity between the policy action and the policy objective, because the central bank reacts to changes in the economic outlook. The stream of literature stemming from the seminal contribution of Romer and Romer (2004) poses that the central bank conducts policy according to an implicit rule

$$\Delta i_t = f(\Omega_t) + \varepsilon_t^{MP}, \tag{10}$$

[15]Examples of monetary policy keywords include “QE”, “pre-commit”, and “interest rate”. Inflation-related keywords include “HICP”, “price pressure”, and “deflation”. Growth-related keywords include “GDP”, “dampen growth”, and “labour market”, while finance-related keywordsinclude “financial markets”, “bond”, and “stock price”.

where changes in the policy instrument i_t are determined by (I) a systematic component $f(\Omega_t)$ depending on the central bank's assessment of the economic outlook Ω_t and (II) a non-systematic monetary policy shock ε_t^{MP}. Since the central bank's assessment of the economic outlook reflects the underlying fundamentals of the economy, a naive regression of inflation (policy objective) on changes in the policy rate (policy instrument) yields biased estimates. Instead, regressing inflation on the non-systematic shock shall retrieve the true impact of monetary policy.

To see this, we first estimate the following local projection

$$\Delta^m p_{t+m} = \alpha^m + \beta^m \Delta i_t + \sum_{j=1}^{24} \gamma_i^m \Delta^j p_{t-j} + \varepsilon_t^m, \tag{11}$$

where $\Delta^m p_{t+m}$ is the m-month ahead change in the log-HICP price index, Δi_t is the change in ECB deposit rate decided at the time t meeting by the Governing Council, and $\sum_{j=1}^{24} \Delta^j p_{t-j}$ is a series of past log-HICP price index changes to control for the dynamics of inflation. Since this specification is not controlling for the economic fundamentals driving both the price level and the policy action, it suffers from omitted variable bias. Moreover, since economic fundamentals are positively correlated with changes in the policy rate through the systematic component, β^m coefficients will be biased upwardly. Indeed, while economic theory predicts a negative relation between the policy rate and inflation, biased point estimates of β^m are often positive and never statistically different from zero (figure 11(a)).

To recover a valid estimates of β^m, one shall replace Δi_t in equation (11) with the non-systematic component ε_t^{MP}, which is uncorrelated with economic fundamentals by definition. Romer and Romer (2004) performed this task assuming that the systematic component is represented by (I) the previous policy rate to account for mean reverting behavior and (II) central bank's forecasts of key macroeconomic indicators

$$\Delta i_t = \alpha + \beta i_{t-1} + \gamma \mathbf{X_t} + \epsilon_t^{MP} \tag{12}$$

Aruoba and Drechsel (2026) improved this specification enlarging the set of macroeconomic indicators with economic information extracted from central bank documents with NLP techniques.

We perform a similar exercise for the ECB, blending the original Romer and Romer (2004) specification with our NLP pipeline. Specifically, we regress changes in the ECB deposit rate on the previous policy rate and our textual tone score index

$$\Delta i_t = \alpha + \beta i_{t-1} + \gamma Score_t + \epsilon_t^{MP} \tag{13}$$

Exploiting the advanced NLP technique employed, our index captures the multi-faceted assessment of the economic outlook, which the Governing Council carries out during each meeting on the base of macroeconomic forecasts and economic indicators. Our estimates confirm the mean reverting behavior of policy and, most importantly, we recover the expected positive relation between the policy instrument and the tone index, suggesting that the central bank raises the deposit rate as economic fundamentals get stronger.

Exploiting the residuals of specification (13), $\hat{\varepsilon}_t^{MP}$, we are finally able to estimate the local projection

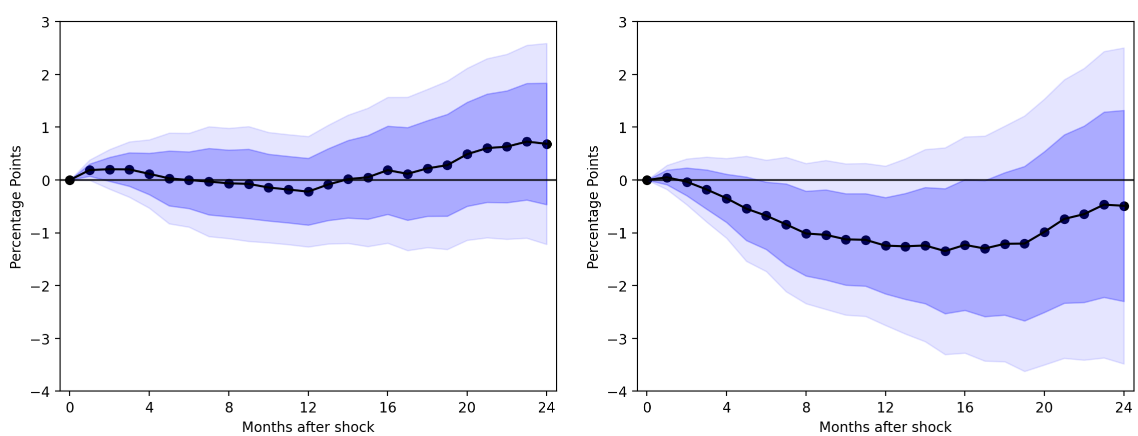
$$\Delta^m p_{t+m} = \alpha^m + \beta^m \hat{\varepsilon}_t^{MP} + \sum_{j=1}^{24} \gamma_i^m \Delta^j p_{t-j} + \varepsilon_t^m, \tag{14}$$

Coefficient	Estimates	Statistics	
α	0.06***	R^2	0.21
i_{t-1}	-0.05***	N. Obs.	264
$Score_t$	0.24***		

Table 2: Estimation results of equation (13). * denotes significance at 10%, ** at 5%, *** at 1% based on HAC standard errors.

which recovers the theoretical negative relation between inflation and the policy instrument (figure 11(b)). Comparing our evidence with both Romer and Romer (2004) and Aruoba and Drechsel (2026) (see (Aruoba and Drechsel, 2026, figure 6)), our estimate of the impact of monetary policy on inflation looks sharper in terms of shape and similar in terms of magnitude. Three remarks are in order: (I) the ECB mandate of price stability, as opposed to FED dual mandate, might explain the greater consistency of our results compared to studies of the US economy, (II) Romer and Romer (2004) and Aruoba and Drechsel (2026) compute inflation based on quarterly GDP deflator since the '80s, while we use the monthly HICP price level because our sentiment index starts from 2000, (III) our estimation sample shrinks as the the number of months ahead increases, possibly explaining the widening of confidence intervals.

According to our estimates, a 25 basis point shock to the policy rate reduces HICP by inflation by 0.28 percentage points ($\beta^{12} \times \Delta i_t = -1.13 \times 0.25 = 0.28$). Overall, these results not only provide an estimate of the impact of ECB monetary policy on inflation, but they also corroborate the validity of our tone score index as credible summary of the ECB Governing Council assessment of the Eurozone economic outlook.



(a) Naive regression

(b) Purged regression

Figure 11: Estimation of equation (11) (panel a) and (14) panel(b). Dots represent point estimates of coefficient β^m, dark (light) areas depict 16-84% (5-95%) confidence intervals

6 Conclusion

In this paper we developed a pipeline based on state-of-the-art NLP techniques to assess the sentiment of monetary policy communication, focusing on ECB press conferences following rate decisions. We show that our methodology provides a meaningful summary of the central bank's assessment of the economic outlook, because our textual tone score index allows to recover a theoretically valid impact of monetary policy on inflation. Future research shall exploit this technology to evaluate the impact of communication on financial markets.

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A Appendix

A.1 Transformer-based Models for Classification

Our NLP pipeline consists of two classification steps: relevance classification and sentiment classification. In the first step, sentences from ECB press conferences are classified according to whether they are relevant to monetary policy. In the second step, the relevant sentences are assigned a sentiment label reflecting their perceived dovish–hawkish tone.

Working with text is challenging, as machines cannot directly understand natural language. Therefore, text must be converted into numerical values that preserve its meaning and structure. These numerical representations are known as *embeddings*, which map words or sentences into vectors of numbers in a continuous space.

Transformer-based models such as BERT, RoBERTa, and GPT are designed to produce high-quality embeddings. During training, these models learn how to encode semantic meaning, capture word order, and model contextual dependencies between tokens. The embeddings produced by Transformer models are *contextual* rather than static, meaning that the representation of a word depends on its surrounding context. For example, the word *"bank"* conveys different meanings in the sentences *"He sat by the river bank"* and *"She deposited money in the bank"*. Transformer models can capture this difference by assigning different embeddings to the word in each sentence.

Given their strong performance in a wide range of downstream NLP tasks, we adopt RoBERTa models in this study. In this appendix, we briefly describe the main components underlying our approach. Appendix A.2 introduces the Transformer architecture, Appendix A.3 discusses BERT, Appendix A.4 describes RoBERTa, and Appendix A.5 explains how Transformer models are fine-tuned for specific tasks.

A.2 The Transformer

Let's first focus on a brief explanation of the Transformer architecture, as described in Vaswani (2017). A Transformer-based model can have an *encoder-decoder*, *encoder-only* or *decoder-only* structure. The model proposed in the aforementioned work explains the canonical encoder-decoder model architecture, displayed in Figure 12. The block on the left is the *encoder*, whose outputs are fed to the *decoder*, the block on the right. Since RoBERTa is based on an *encoder-only* Transformer architecture, we focus our discussion on the encoder component.

An encoder block is composed of a stack of L identical layers, 12 for the `BERT-base` model and 24 for the `BERT-large` model. "Each layer has two sub-layers. The first is a *Multi-Head self-Attention mechanism* (MHA), and the second is a simple, position-wise *fully connected feed-forward network* $g(\cdot)$. We employ a *residual connection*[16] around each of the two sub-layers, followed by

[16] A *residual or skip connection* connects the output of a previous layer with the output of a future layer, skipping any transformation in between. In the context of a Transformer encoder layer, with the first residual connection, the input to the first sub-layer X connects to the output of the first sub-layer MHA(X), and with the second residual connection, the output of the first sub-layer, after applying layer normalization, connects to the output of the second sub-layer $g(\mathrm{H})$.

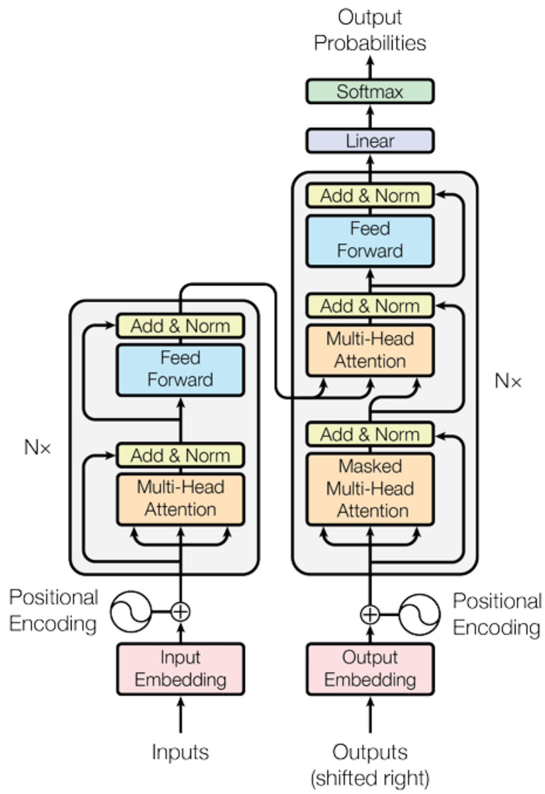

Figure 12: The Transformer - model architecture. Source: Vaswani (2017).

layer normalization[17]" Vaswani (2017), as shown in the equations below

$$
\begin{aligned}
\mathbf{H} &= \mathrm{MHA}(\mathbf{X}) \\
\mathbf{H} &= \mathrm{LayerNorm}(\mathbf{H} + \mathbf{X}) \\
\mathbf{F} &= g(\mathbf{H}) \\
\mathbf{H} &= \mathrm{LayerNorm}(\mathbf{F} + \mathbf{H}).
\end{aligned}
\tag{15}
$$

The core of the Transformer model is the *self-attention mechanism*, which allows the model to

[17] *Layer normalization* is a variant of batch normalization that normalizes the inputs across the features for each data sample rather than across the batch. First, we normalize the input features

$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

where μ and σ^2 are respectively the mean and variance across the feature dimension, and ϵ is a small constant for numerical stability. Secondly, we apply the learned scale α and shift β parameters to $\hat{x}_i$ and obtain

$$\mathrm{output}_i = \alpha \cdot \hat{x}_i + \beta$$

focus on different parts of the input sequence, mimicking the way human attention works. This mechanism works by computing *attention scores* between pairs of tokens $(\mathbf{x_i}, \mathbf{x_j})$ in the sequence. Specifically, the *scaled dot-product attention* function $\alpha(\cdot, \cdot)$ is used

$$\alpha(\mathbf{x_i}, \mathbf{x_j}) = \frac{1}{\sqrt{d}} \mathbf{x_i}^\intercal \mathbf{x_j}, \tag{16}$$

where $\mathbf{x_i}$ and $\mathbf{x_j}$ are two d-dimensional vectors. Now, we can define the *self-attention layer* (for a single token i) as

$$\mathbf{h_i} = \sum_{j=1}^{n} \text{Softmax}_j \left(\frac{1}{\sqrt{d}} \mathbf{x_i}^\intercal \mathbf{x_j} \right) \mathbf{x_j}, \tag{17}$$

where the Softmax function ensures that the attention scores are normalized, converting them into probabilities, also called *attention weights*. This formulation of the self-attention layer lacks trainable parameters. Hence, we must introduce 3 *trainable matrices*, namely $\mathbf{W}_q \in \mathbb{R}^{d \times d_k}$, $\mathbf{W}_k \in \mathbb{R}^{d \times d_k}$, $\mathbf{W}_v \in \mathbb{R}^{d \times d_v}$, such that we can define the following vectors

$$\mathbf{q_i} = \mathbf{W}_q^\intercal \cdot \mathbf{x_i},\ \mathbf{k_j} = \mathbf{W}_k^\intercal \cdot \mathbf{x_j},\ \mathbf{v_j} = \mathbf{W}_v^\intercal \cdot \mathbf{x_j}, \tag{18}$$

which we call *query*, *key* and *value*. We substitute vectors $\mathbf{q_i}, \mathbf{k_i}$ and $\mathbf{v_i}$ in equation (17) and obtain

$$\mathbf{h_i} = \sum_{j=1}^{n} \text{Softmax}_j \left(\frac{1}{\sqrt{d_k}} \mathbf{q_i}^\intercal \mathbf{k_j} \right) \mathbf{v_j}, \tag{19}$$

where the attention scores are calculated based on the dot product of query and key vectors, scaled by the square root of their dimension. Finally, we can rewrite the self-attention layer in vectorized form by stacking the n input vectors $\mathbf{x_i}$ into a matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$.

$$\text{Attention Scores} = \frac{\mathbf{Q}\mathbf{K}^\intercal}{\sqrt{d_k}} \in \mathbb{R}^{n \times n} \tag{20}$$

$$\text{Attention Weights} = \text{Softmax}\left(\text{Attention Scores}\right) \in \mathbb{R}^{n \times n} \tag{21}$$

$$\mathbf{H} = \text{Attention Weights} \cdot \mathbf{V} \in \mathbb{R}^{n \times d_v}, \tag{22}$$

where $\mathbf{H}$ is the output of the self-attention layer, $\mathbf{Q} = \mathbf{X}\mathbf{W}_q \in \mathbb{R}^{n \times d_k}$, $\mathbf{K} = \mathbf{X}\mathbf{W}_k \in \mathbb{R}^{n \times d_k}$ and $\mathbf{V} = \mathbf{X}\mathbf{W}_v \in \mathbb{R}^{n \times d_v}$.

To enhance the model's capability, Transformers employ *Multi-Head Attention* (MHA), which allows the model to capture various aspects of the input sequence simultaneously, e.g. different meanings of a word in different contexts. The MHA layer works by computing $t = 1, \ldots, h$ separate sets of keys, queries and values, with h being the number of *attention heads*

$$\mathbf{Q}_t = \mathbf{X}\mathbf{W}_{q,t},\ \mathbf{K}_t = \mathbf{X}\mathbf{W}_{k,t},\ \mathbf{V}_t = \mathbf{X}\mathbf{W}_{v,t} \tag{23}$$

$$\mathbf{H}_t = \text{Softmax}\left(\frac{\mathbf{Q}_t \mathbf{K}_t^\intercal}{\sqrt{d_k}} \right) \mathbf{V}_t. \tag{24}$$

The outputs from all attention heads $\mathbf{H}_t$ are concatenated over the embedding dimension and projected again using the *trainable output matrix* $\mathbf{W}_o \in \mathbb{R}^{hd_v \times d_o}$, where d_o is the output size

$$\text{MHA}(\mathbf{X}) = [\mathbf{H}_1, \ldots, \mathbf{H}_h] \cdot \mathbf{W}_o \in \mathbb{R}^{n \times d_o}. \tag{25}$$

Figure 12 highlights another key component of Transformer models known as *positional encodings or embeddings*. These positional encodings, represented by the matrix $\mathbf{E}$, are either concatenated or summed with the token embeddings $\mathbf{Y}$ before being fed into the first Multi-Head Attention (MHA) layer. The purpose of positional encodings is to provide information about the position of each token within the input sequence, which allows the model to capture the order of tokens. The Transformer model uses sinusoidal functions to generate positional embeddings. The idea is to alternate sine and cosine functions of the same frequency w_j

$$w_j = \frac{1}{10000^{\frac{2j}{d}}}, \tag{26}$$

where d is the dimension of both the positional encoding and token embedding vectors. For position i and dimension j we get

$$[\mathbf{E}]_{i,2j} = sin\left(w_j\right), \; [\mathbf{E}]_{i,2j+1} = cos\left(w_j\right). \tag{27}$$

The positional embeddings are then added to the token embeddings to obtain the input matrix $\mathbf{X}$

$$[\mathbf{X}]_{i,:} = [\mathbf{Y}]_{i,:} + [\mathbf{E}]_{i,:}, \tag{28}$$

where each row $[\mathbf{E}]_{i,:}$ uniquely encodes the position of every element in the sequence. This combined representation incorporates the content of the token and its position in the sequence.

A.3 BERT

BERT (*Bidirectional Encoder Representation from Transformers*) is a multi-layer bidirectional Transformer encoder "designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers" Devlin (2018). The usage of BERT involves two main steps: *pre-training* and *fine-tuning*. As described in Section 4, our relevance and sentiment text classification require pre-trained Transformer-based models with BERT architecture. We adapted these models to our specific task through fine-tuning. In this Section, however, we will explain the pre-training procedure for BERT, which enables the model to learn general language representations from extensive text corpora. BERT is pre-trained via two unsupervised tasks: *Masked Language Modeling* (MLM) and *Next Sentence Prediction* (NSP).

MLM enables BERT to transition from unidirectional to bidirectional representations. This method involves *masking* certain tokens in the input text and then training the model to predict these masked tokens based on the context provided by the remaining words in the sentence. Unlike traditional language models that predict the next word in a sequence (which is unidirectional), Masked LM allows BERT to utilize context from both directions (left and right) around a masked token. In the BERT paper, approximately 15% of the tokens in each input sequence are masked. "If the i-th token is chosen, we replace the i-th token with (1) the `[MASK]` token 80% of the time (2) a random token 10% of the time (3) the unchanged i-th token 10% of the time[18]. Then, T_i[19] will

[18] To bias the representation towards the actual observed word.

[19] The final hidden representation for the *i*-th input token.

be used to predict the original token with cross-entropy loss” Devlin (2018). Let’s suppose that the unlabeled sentence is “*i like this book*”, and during the random masking procedure, we choose the 4-th token. The masking procedure can be illustrated with the following example:

- 80% of the time we replace token “*book*” with `[MASK]`.

  “i like this book” ⟶ *“i like this* `[MASK]`*”*

- 10% of the time we replace token “*book*” with a random word, e.g. “*apple*”.

  “i like this book” ⟶ *“i like this apple”*

- 10% of the time we keep the 4-th token unchanged.

  “i like this book” ⟶ *“i like this book”*

On the other hand, NSP helps BERT learn the relationships between sentences, a capability not captured by language modeling alone but essential for tasks such as Question Answering (QA) and Natural Language Inference (NLI). In NSP, the two sentences are separated by a special token `[SEP]` and preceded by the `[CLS]` token used for classification tasks

Input: `[CLS]` Sentence A `[SEP]` Sentence B `[SEP]`.

For each sentence pair (Sentence A, Sentence B) (1) 50% of the time *Sentence B* is the actual sentence that follows *Sentence A* in the text (labeled as *“IsNext”*), (2) the remaining 50% of the time *Sentence B* is chosen at random from the corpus (labeled ad *“NotNext”*). For instance,

(1) Sentence B: *“he bought a gallon* `[MASK]` *milk”* is the true sentence that follows Sentence A: *“the man went to* `[MASK]` *store”*. The label is *“IsNext”*.

(2) Sentence B: *“my dog* `[MASK]` *barking”* is a random sentence from the corpus. The label is *“NotNext”*.

A.4 RoBERTa

RoBERTa (*Robustly optimized BERT approach*) is an improved version of BERT, sharing the same architecture but trained with several improvements:

(1) *Extended training.* RoBERTa was trained for significantly more steps (125M or 31M steps vs. BERT’s 1M steps), using larger batch sizes (2000 or 8000 vs. 256) and over more data (160GB, about 10 times more than for BERT).

(2) *Removal of the Next Sentence Prediction (NSP) objective.* RoBERTa was trained exclusively on the Masked Language Modeling (MLM) objective, as its authors found that eliminating NSP either matched or slightly improved performance on downstream tasks.

(3) *Training on longer sequences.* RoBERTa was trained with sequences of up to 512 tokens, while BERT was trained mostly on shorter sequences of 128 tokens.

(4) *Dynamic masking.* Unlike BERT's static masking (where mask tokens are fixed during data preprocessing), RoBERTa uses dynamic masking, meaning that sequences are masked differently throughout the training process. Each training sequence is masked in 10 different ways over the 40 epochs of training.

We use both the base and large variants of RoBERTa, whose architectural characteristics are reported in Table 3. The base model has 12 layers and fewer parameters, while the large model has higher capacity and is therefore better suited for more complex tasks such as sentiment classification compared to relevance classification.

Model	L	H	N	Number of Parameters	Vocabulary size
RoBERTa-base	12	12	768	∼125M	50,265
RoBERTa-large	24	16	1024	∼355M	50,265

Table 3: Architecture characteristics of the RoBERTa models. L is the number of layers, H is the hidden dimension, N is the maximum input sequence length and Number of parameters refers to the number of the embedding and Transformer model parameters.

A.5 Fine-Tuning Pre-Trained Transformer-based Models

Training a deep neural network typically requires a large amount of data. When such extensive datasets are unavailable, we can leverage pre-trained models (e.g. Transformers) and adapt them for our specific task. This approach is known as *Transfer Learning.* As the name suggests, rather than starting from scratch, Transfer Learning involves transferring the knowledge acquired by a large, pre-trained model and applying it to a different but related problem. As shown in Figure 13, the process begins with a pre-trained model $M_{\mathcal{D}}$ (e.g. RoBERTa), which has already been trained on a large general-purpose dataset $\mathcal{D}$ (e.g. as described in Section A.3). We can then adapt this model to a new task t using a task-specific dataset $\mathcal{D}_t$, creating a new model $M_{\mathcal{D}_t}$. This process can be repeated multiple times across different tasks and datasets.

The simplest and most common way to perform Transfer Learning is through *Fine-Tuning.* Fine-tuning adapts a pre-trained model to a new task by *re-training some of its weights* using a task-specific dataset—in our case, the corpus of ECB press conference texts—to improve its performance on that specific task. Typically, the final layer of the model is replaced with a new layer that suits the new task. For example, we might add a *classification head* with randomly initialized weights that generate probabilities for the target classes.

A Transformer model is composed of multiple layers, each capturing different levels of information. The earlier layers focus on extracting general, low-level features, while the later layers extract more task-specific, high-level features. Thus, during fine-tuning, it is common to "freeze" the earlier layers, leaving only the later layers trainable (as shown in Figure 15). This allows the model to adapt to the new task without losing the general knowledge it gained during pre-training. Fine-tuning is usually done with a low learning rate to avoid wiping out all the pre-trained knowledge[20].

Building on this framework, Figure 16 illustrates how RoBERTa is adapted for the sentence-level sentiment classification task in our setting. Each context-augmented sentence is fed into the

[20]This phenomenon is known as *catastophic forgetting.*

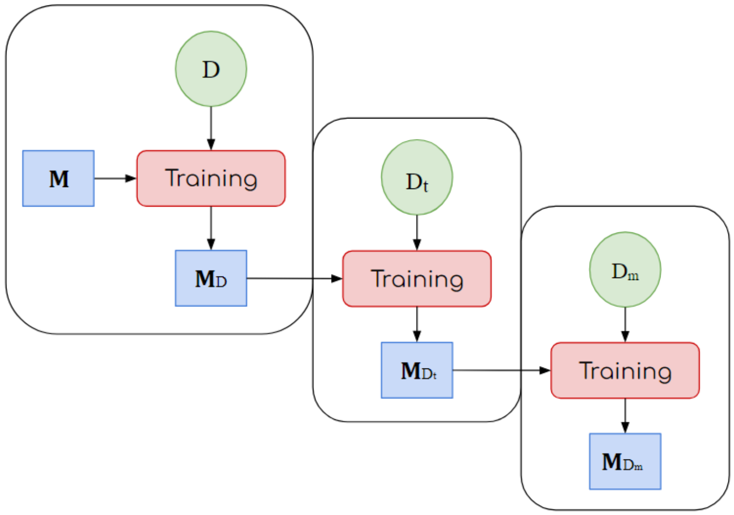

Figure 13: Transfer Learning.

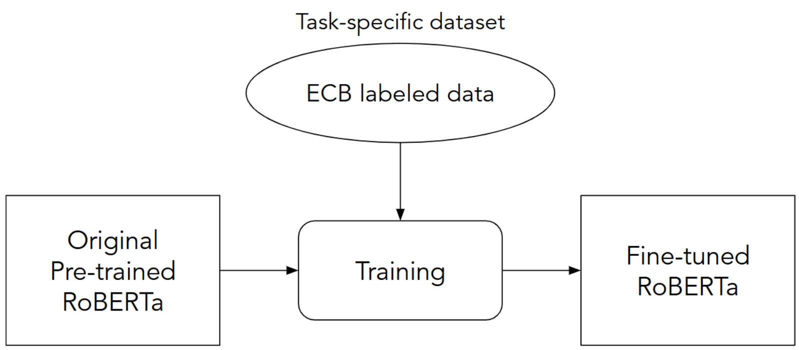

Figure 14: Adapting a Transformer-based model to the ECB dataset through fine-tuning.

pre-trained RoBERTa model, which is subsequently fine-tuned to specialize the general-purpose language representations for monetary policy sentiment classification. During fine-tuning, only a subset of the model parameters is updated, typically corresponding to the last l Transformer layers, while the remaining layers are kept frozen to preserve the general linguistic knowledge learned during pre-training. The model processes the input sentence by generating contextualized embeddings for each token, where the representation of each token depends on the full sentence context. The hidden representation of the special classification token $\langle s \rangle$[21] from the final Transformer layer is used as a fixed-length sentence representation. To perform classification, a task-specific five-class *classification head* is added on top of the model. This head consists of a dense layer with a non-linear activation, followed by a dropout layer and an output projection layer that maps the sentence

[21] This token signals the start of the sentence.

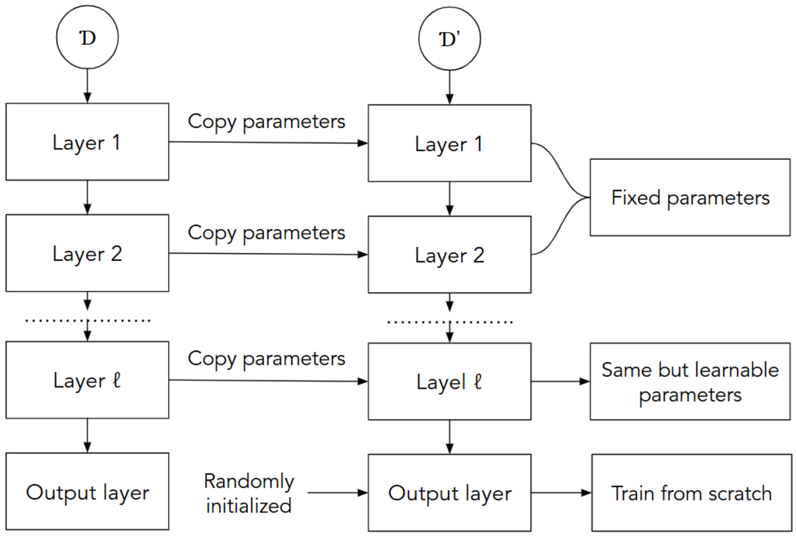

Figure 15: Fine-Tuning.

representation to class logits[22]. The parameters of the classification head are randomly initialized and trained from scratch during fine-tuning. The head outputs a probability distribution over the five sentiment classes, and the class with the highest predicted probability is selected as the final sentiment label for the sentence.

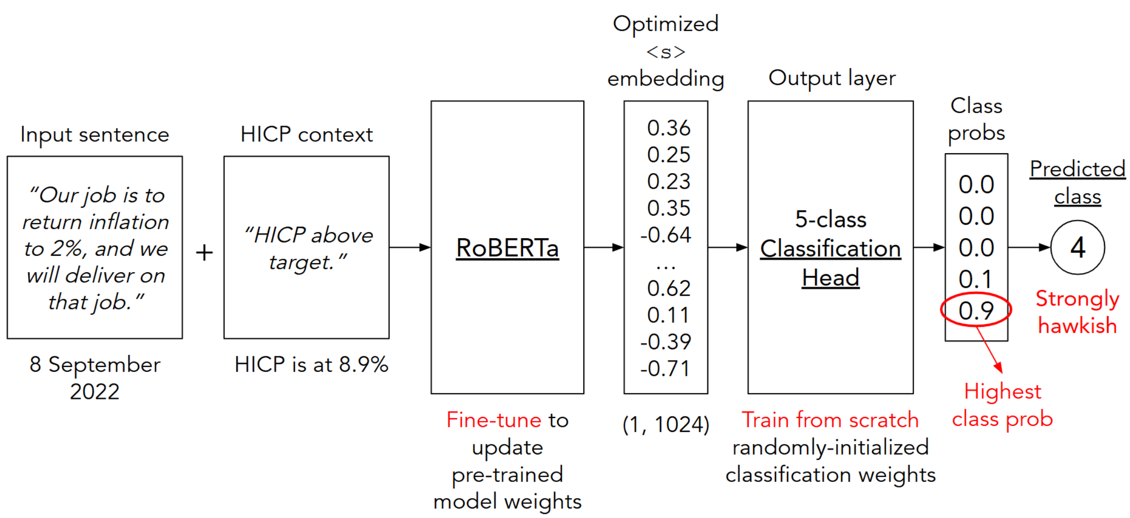

Figure 16: Fine-tuning RoBERTa for classification: an example.

[22]Raw, un-normalized scores output by the model before being transformed into a probability distribution via the softmax function.

B First-pass Relevance Filter

This appendix lists the keywords used to perform the first-pass relevance filter of section 4.1.

Category	**Terms**
Connectors	accordingly, although, and, at the same time, but, consequently, furthermore, hence, however, in addition, indeed, instead, moreover, nevertheless, nonetheless, on the other hand, rather, subsequently, therefore, though, yet
Common	celebrate, further information, good afternoon, happy new year, i will now outline, i will now report, ladies and gentlemen, let me now, merry christmas, please, press release, publish, scheduled, the governing council today decided, to be held, we decided to, website, will be released
Questions	a question on, add-on question, ask you a question, could you, do you have, do you think, does that, first question is on, follow-up question, i am wondering, i have a question, i have a quick question, I have a two-pronged question, i have got a quick question, i have one question, i have three questions, i have two questions, i just wanted to ask, i want to ask, i wanted to ask, i wonder whether, i would like to ask, i've got a question, i've got two questions, if i may, if you please, in your opinion, my first question, my question, my second question, my third question, one last question, one question is, question:, second question is on, that is the first question, the first one is on, the first question is, the first question was, the next question is, the next question regards, the other question is, the other question regards, the question is, the second one is on, the second question is, the second question was, the third question is, the third question was, what do you, what is your, what would, would you, you have mentioned, you mentioned, you said, you were saying, your last press conference, your press conference
Economic	accommodation, accommodative, adverse factor, anchor, APP, balance sheet, bank, bank rate, challenge, consumption, credit, crisis, CSPP, cut, cycle, cyclical, data, decelerate, decline, decrease, deficit, deflation, demand, deposit, disinflation, disrupt, dove, downward, drop, ease, easing, economic, economic activity, economy, employment, exchange rate, expand, export, fall, fell, financial, financial sector, financing, fiscal, forecast, forward guidance, fund rate, GDP, good news, good place, grow, growth, hawk, headline, HICP, high, hike, hold, improve, increase, inflation, inflation expectation, interest rate, inventor, investment, job market, job retention, labor, labour, lending, liquid, liquidity, loan, low, market, monetary, monetary policies, monetary policy, monetary policy transmission, negative, non-accelerating, outlook, pause, pausing, PEPP, pessimistic, PMI, pmi indicator, portfolio, positive, pressure, price, productivity, projection, QE, QT, quantitative easing, quantitative tightening, raise, rapid, rate, rate path, real sector, rebound, recovery, reduce, refinancing, reinvest, repay, required, reserve, resilient, restrictive, rise, rising, risk, savings, services, slack, slow, sluggish, stable, stagflation, stimulus, strength, strong, subdue, supply chain, tariff, tighten, trade, uncertainty, unemployment, upward, urgent, wage, weak, well positioned, well-positioned, zero lower bound

Table 4: Economics-related keywords used for relevance filtering in Step 1.

C Data Labeling Examples

All sentences in the dataset were independently annotated by two annotators. Each annotator assigned one of six labels to every sentence, reflecting the monetary policy stance indicated. In cases of disagreement, a final label was jointly agreed upon. The labels are defined as follows:

- *Label -2 (Strongly Dovish):* Sentences that explicitly indicate a monetary policy action aimed at stimulating the economy and/or raising inflation, such as lowering interest rates or expanding the central bank's balance sheet. These statements reflect a strongly accommodative policy stance with a clear intention to act.
- *Label -1 (Dovish):* sentences that describe weak economic conditions, such as slow growth, high unemployment, or inflation below target. These reflect negative economic considerations but do not indicate a concrete policy action.
- *Label 0 (Neutral):* Sentences that are neither dovish nor hawkish, providing general information without suggesting a direction for monetary policy.
- *Label 1 (Hawkish):* Sentences that highlight strong economic conditions, rising inflation, or low unemployment. These reflect positive economic considerations but do not explicitly signal a tightening action.
- *Label 2 (Strongly Hawkish):* Sentences that explicitly indicate a tightening of monetary policy, such as raising interest rates, in response to high inflation or an economy growing faster than desired. These statements reflect a strongly restrictive policy stance with a clear intention to act.
- *Label 3 (Irrelevant):* Sentences that do not pertain to monetary policy and are included for training the relevance classification model.

The following tables contain 10 sentence labeling examples per category. Each sentence is accompanied by a *context* label indicating whether inflation was *below target*, *around target*, or *above target* at the time of the communication. This contextual information is particularly useful when annotating ambiguous sentences, as the interpretation of the tone (dovish, hawkish, neutral) can depend on the prevailing inflationary environment.

Table 5: Strongly dovish examples.

Sentence	**Context**
Looking ahead, our monetary policy stance will remain accommodative for as long as necessary, in line with the forward guidance provided in July.	HICP below target
In any case, we will conduct net asset purchases under the PEPP until the Governing Council judges that the coronavirus crisis phase is over.	HICP below target
Against this overall background, our monetary policy stance will remain accommodative for as long as needed.	HICP below target

Sentence	Context
This procedure will also remain in use for the Eurosystem's special-term refinancing operations with a maturity of one maintenance period, which will continue to be conducted for as long as needed, and at least until the end of the second quarter of 2014.	HICP below target
So, purchases are intended to run until the end of September 2016, and in any case until we see a sustained adjustment in the path of inflation that is consistent with our aim of achieving inflation rates over etc etc, and then there is this new sentence explaining exactly what we mean by sustained and by medium-term.	HICP below target
We will continue our purchases under the pandemic emergency purchase programme (PEPP) with a total envelope of €1,350 billion.	HICP below target
The mandate is price stability; And the Governing Council is unanimous and determined to take actions, any action, within this mandate to reach the objective of price stability.	HICP below target
Looking ahead, the Governing Council is strongly determined to safeguard this anchoring.	HICP below target
The Governing Council is unanimous in its commitment to using also unconventional instruments within its mandate should it become necessary to further address risks of too prolonged a period of low inflation.	HICP below target
On the basis of this updated assessment, the Governing Council will recalibrate its instruments, as appropriate, to respond to the unfolding situation and to ensure that financing conditions remain favourable to support the economic recovery and counteract the negative impact of the pandemic on the projected inflation path.	HICP below target

Table 6: Dovish examples.

Sentence	Context
The risks surrounding the economic outlook for the euro area remain on the downside.	HICP below target
Underlying price pressures in the euro area are expected to remain subdued over the medium term.	HICP below target
As I said in the introductory statement, HICP inflation has gone down considerably.	HICP below target
In the view of the Governing Council, the economic outlook is subject to increased downside risks, mainly stemming from a scenario of ongoing financial market tensions affecting the real economy more adversely than currently foreseen.	HICP above target
The most recent data clearly confirm that economic activity in the euro area is weakening, with contracting domestic demand and tighter financing conditions.	HICP above target

Sentence	Context
Incoming information signals that the euro area economic recovery is losing momentum more rapidly than expected, after a strong, yet partial and uneven, rebound in economic activity over the summer months.	HICP below target
The inflation numbers that we have for September, for August are in negative territory, -0.2% followed by -0.3%.	HICP below target
Overall, credit dynamics remain weak.	HICP above target
Overall, labour market conditions remain weak.	HICP below target
As I said in the introductory statement, HICP inflation has gone down considerably.	HICP below target

Table 7: Neutral examples.

Sentence	Context
Medium to long-term inflation expectations continue to be firmly anchored in line with price stability.	HICP below target
The new Eurosystem staff projections show headline inflation averaging 2.1% in 2025, 1.9% in 2026, 1.8% in 2027 and 2.0% in 2028.	HICP above target
The rate on the deposit facility will remain unchanged at 0.00%.	HICP below target
Second, we are closely monitoring money market conditions and their potential impact on our monetary policy stance and its transmission to the economy.	HICP below target
Inflation expectations for the euro area over the medium to long term continue to be firmly anchored in line with our aim of maintaining inflation rates below, but close to, 2%.	HICP below target
We are not pre-committing to a particular rate path.	HICP above target
Measures of longer-term inflation expectations have remained broadly stable, with most standing at around 2 per cent.	HICP above target
Inflation is currently at around our two per cent medium-term target.	HICP at target
So we are well-positioned to navigate those circumstances.	HICP at target
We think that our monetary policy is perfectly adequate with respect to our stance.	HICP above target

Table 8: Hawkish examples.

Sentence	Context
The risks to the inflation outlook continue to be on the upside and have intensified, particularly in the short term.	HICP above target
To sum up, annual inflation rates are projected to remain elevated in 2006 and 2007, with risks to this outlook remaining clearly on the upside.	HICP at target
We can see improvements in the financial markets.	HICP below target

Sentence	Context
With regard to price developments, annual HICP inflation has remained considerably above the level consistent with price stability since last autumn, standing at 3.6% in September according to Eurostat's flash estimate, after 3.8% in August.	HICP above target
Inflation rose to 9.9 per cent in September, reflecting further increases in all components.	HICP above target.
Looking further ahead, the conditions remain in place for ongoing economic expansion in the euro area.	HICP at target
Measures of underlying inflation have thus remained at elevated levels.	HICP above target
We also see wages continuing to increase in the future.	HICP above target
Looking ahead, we remain confident that the recovery of economic activity will continue.	HICP above target.
Past increases in energy costs are still pushing up prices across the economy.	HICP above target

Table 9: Strongly hawkish examples.

Sentence	Context
Our future decisions will ensure that the key ECB interest rates will be set at sufficiently restrictive levels for as long as necessary to achieve a timely return of inflation to our two per cent medium-term target.	HICP above target
We took today's decision, and expect to raise interest rates further, to ensure the timely return of inflation to our two per cent medium-term inflation target.	HICP above target
Against this background, it is imperative to ensure that medium to longer-term inflation expectations remain firmly anchored at levels in line with price stability.	HICP above target
Summing up, today we have raised the three key ECB interest rates by 75 basis points, and expect to raise interest rates further, to ensure the timely return of inflation to our medium-term target.	HICP above target
We will keep policy rates sufficiently restrictive for as long as necessary to achieve this aim.	HICP above target
Well, that's what we say very clearly in the monetary policy statement: that we will stay in restrictive territory as long as it's necessary to reach our 2% target; And that is clearly the case.	HICP above target
We will have further rate increases in the future so the normalisation process continues.	HICP above target
As I have said, the rate will be the one that delivers the 2% medium-term target that we have in order to deliver price stability.	HICP above target

Sentence	Context
Our future decisions will ensure that the key ECB interest rates will be brought to levels sufficiently restrictive to achieve a timely return of inflation to our two per cent medium-term target and will be kept at those levels for as long as necessary.	HICP above target

Table 10: Irrelevant examples.

Sentence	Context
To maintain the credibility of the fiscal policy framework it is essential to fully abide by its rules and to implement them in every respect.	HICP at target
National fiscal and structural policies should aim at making the economy more productive and competitive, which would help to raise potential growth and reduce price pressures in the medium term.	HICP above target
Such structural reforms should target improvements in competitiveness and adjustment capacities, as well as aim to increase sustainable growth and employment.	HICP below target
Full and consistent implementation of the Pact is key for confidence in our fiscal framework.	HICP below target
It was in particular, with my colleague, the Governor of the Belgian central bank, to those who were directly involved in this particular matter.	HICP above target
On the stimulus front, we are quite encouraged to have seen for the first time in many years fiscal and monetary work together.	HICP below target
We welcome the European Commission's recent guidance calling for EU Member States to strengthen fiscal sustainability and the Eurogroup's statement on the fiscal stance for the euro area in 2025.	HICP above target
So this does not come as a surprise to us.	HICP above target
I really hope that all Croatian economic actors will feel compelled to actually respect those requirements, which I know are embedded currently now in the draft laws that will be implemented.	HICP above target